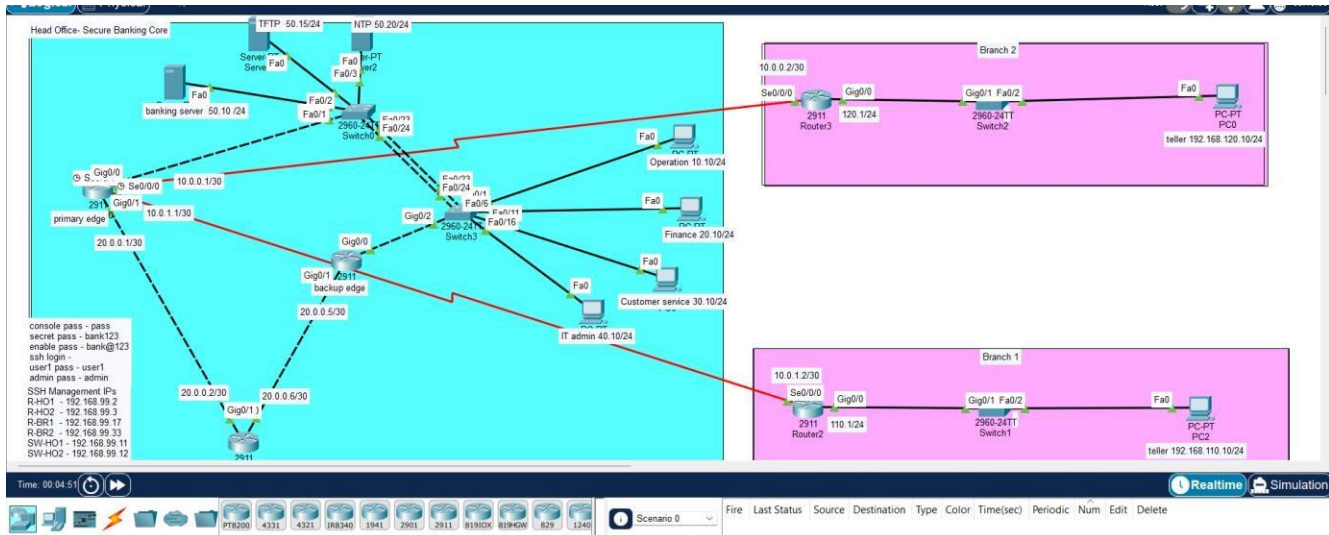


SECURE BANKING HEAD OFFICE s BRANCH NETWORK

Implementation • Connectivity • Security • Failover • Troubleshooting
Validation Report



1. Interface & Physical Connectivity Validation

Purpose: Verify that all connected interfaces are administratively and operationally up.

Test / Evidence	Command or Action	Expected Result
Router interfaces	show ip interface brief	Required interfaces are up/up
Switch ports	show interfaces status	Connected ports show connected

Evidence Screenshot

Router Interfaces

R-HO1

```

Password:
R-HO1#sh ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES manual up          up
GigabitEthernet0/0.10   192.168.10.2    YES manual up          up
GigabitEthernet0/0.20   192.168.20.2    YES manual up          up
GigabitEthernet0/0.30   192.168.30.2    YES manual up          up
GigabitEthernet0/0.40   192.168.40.2    YES manual up          up
GigabitEthernet0/0.50   192.168.50.2    YES manual up          up
GigabitEthernet0/0.99   192.168.99.2    YES manual up          up
GigabitEthernet0/1      20.0.0.1        YES manual up          up
GigabitEthernet0/2      unassigned      YES unset  administratively down down
Serial0/0/0           10.0.1.1        YES manual up          up
Serial0/0/1           10.0.0.1        YES manual up          up
Vlan1                 unassigned      YES unset  administratively down down
R-HO1#

```

R-HO2

```

R-HO2#sh ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0      unassigned      YES unset  up          up
GigabitEthernet0/0.10   192.168.10.3    YES manual up          up
GigabitEthernet0/0.20   192.168.20.3    YES manual up          up
GigabitEthernet0/0.30   192.168.30.3    YES manual up          up
GigabitEthernet0/0.40   192.168.40.3    YES manual up          up
GigabitEthernet0/0.50   192.168.50.3    YES manual up          up
GigabitEthernet0/0.99   192.168.99.3    YES manual up          up
GigabitEthernet0/1      20.0.0.5        YES manual up          up
GigabitEthernet0/2      unassigned      YES unset  administratively down down
Vlan1           unassigned      YES unset  administratively down down
R-HO2#

```

R-BR1

```
R-BR1#sh ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES manual up          up
GigabitEthernet0/0.99 192.168.99.17   YES manual up          up
GigabitEthernet0/0.110 192.168.110.1   YES manual up          up
GigabitEthernet0/1    unassigned      YES unset administratively down down
GigabitEthernet0/2    unassigned      YES unset administratively down down
Serial0/0/0          10.0.1.2        YES manual up          up
Serial0/0/1          unassigned      YES unset administratively down down
Vlan1              unassigned      YES unset administratively down down
R-BR1#
```

R-BR2

```

R-BR2#sh ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES manual up          up
GigabitEthernet0/0.99 192.168.99.33   YES manual up          up
GigabitEthernet0/0.120 192.168.120.1   YES manual up          up
GigabitEthernet0/1    unassigned      YES unset administratively down down
GigabitEthernet0/2    unassigned      YES unset administratively down down
Serial0/0/0          10.0.0.2        YES manual up          up
Serial0/0/1          unassigned      YES unset administratively down down
Vlan1              unassigned      YES unset administratively down down
R-BR2#

```

ISP Router

```
ISP>en
ISP#sh ip int brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    20.0.0.2        YES manual up          up
GigabitEthernet0/1    20.0.0.6        YES manual up          up
GigabitEthernet0/2    unassigned      YES manual down         down
Vlan1              unassigned      YES unset administratively down down
ISP#
```

Switch Ports

SW-HO1

```
SW-HO1#sh int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	trunk	a-full	a-100	
Fa0/1		connected	50	a-full	a-100	10/100BaseTX
Fa0/2		connected	50	a-full	a-100	10/100BaseTX
Fa0/3		connected	50	a-full	a-100	10/100BaseTX
Fa0/4		notconnect	50	auto	auto	10/100BaseTX
Fa0/5		notconnect	50	auto	auto	10/100BaseTX
Fa0/6		notconnect	1	auto	auto	10/100BaseTX
Fa0/7		notconnect	1	auto	auto	10/100BaseTX
Fa0/8		notconnect	1	auto	auto	10/100BaseTX
Fa0/9		notconnect	1	auto	auto	10/100BaseTX
Fa0/10		notconnect	1	auto	auto	10/100BaseTX
Fa0/11		notconnect	1	auto	auto	10/100BaseTX
Fa0/12		notconnect	1	auto	auto	10/100BaseTX
Fa0/13		notconnect	1	auto	auto	10/100BaseTX
Fa0/14		notconnect	1	auto	auto	10/100BaseTX
Fa0/15		notconnect	1	auto	auto	10/100BaseTX
Fa0/16		notconnect	1	auto	auto	10/100BaseTX
Fa0/17		notconnect	1	auto	auto	10/100BaseTX
Fa0/18		notconnect	1	auto	auto	10/100BaseTX
Fa0/19		notconnect	1	auto	auto	10/100BaseTX
Fa0/20		notconnect	1	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		connected	trunk	a-full	a-100	10/100BaseTX
Fa0/24		connected	trunk	a-full	a-100	10/100BaseTX
Gig0/1		connected	trunk	a-full	a-100	10/100/1000BaseTX
Gig0/2		notconnect	1	auto	auto	10/100/1000BaseTX

```
SW-HO1#
```

SW-HO2

```
SW-HO2#sh int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Pol		connected	trunk	a-full	a-100	
Fa0/1		connected	10	a-full	a-100	10/100BaseTX
Fa0/2		notconnect	10	auto	auto	10/100BaseTX
Fa0/3		notconnect	10	auto	auto	10/100BaseTX
Fa0/4		notconnect	10	auto	auto	10/100BaseTX
Fa0/5		notconnect	10	auto	auto	10/100BaseTX
Fa0/6		connected	20	a-full	a-100	10/100BaseTX
Fa0/7		notconnect	20	auto	auto	10/100BaseTX
Fa0/8		notconnect	20	auto	auto	10/100BaseTX
Fa0/9		notconnect	20	auto	auto	10/100BaseTX
Fa0/10		notconnect	20	auto	auto	10/100BaseTX
Fa0/11		connected	30	a-full	a-100	10/100BaseTX
Fa0/12		notconnect	30	auto	auto	10/100BaseTX
Fa0/13		notconnect	30	auto	auto	10/100BaseTX
Fa0/14		notconnect	30	auto	auto	10/100BaseTX
Fa0/15		notconnect	30	auto	auto	10/100BaseTX
Fa0/16		connected	40	a-full	a-100	10/100BaseTX
Fa0/17		notconnect	40	auto	auto	10/100BaseTX
Fa0/18		notconnect	40	auto	auto	10/100BaseTX
Fa0/19		notconnect	40	auto	auto	10/100BaseTX
Fa0/20		notconnect	40	auto	auto	10/100BaseTX
Fa0/21		notconnect	1	auto	auto	10/100BaseTX
Fa0/22		notconnect	1	auto	auto	10/100BaseTX
Fa0/23		connected	trunk	a-full	a-100	10/100BaseTX
Fa0/24		connected	trunk	a-full	a-100	10/100BaseTX
Gig0/1		notconnect	1	auto	auto	10/100/1000BaseTX
Gig0/2		connected	trunk	a-full	a-100	10/100/1000BaseTX

```
SW-HO2#
```

SW-BR1

```
SW-BR1#sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Name      notconnect  110       auto    auto   10/100BaseTX
Fa0/2     connected  110       a-full    a-100   10/100BaseTX
Fa0/3     notconnect 110       auto     auto    10/100BaseTX
Fa0/4     notconnect 110       auto     auto    10/100BaseTX
Fa0/5     notconnect 110       auto     auto    10/100BaseTX
Fa0/6     notconnect 110       auto     auto    10/100BaseTX
Fa0/7     notconnect 110       auto     auto    10/100BaseTX
Fa0/8     notconnect 110       auto     auto    10/100BaseTX
Fa0/9     notconnect 110       auto     auto    10/100BaseTX
Fa0/10    notconnect 110       auto     auto    10/100BaseTX
Fa0/11    notconnect 1         auto     auto    10/100BaseTX
Fa0/12    notconnect 1         auto     auto    10/100BaseTX
Fa0/13    notconnect 1         auto     auto    10/100BaseTX
Fa0/14    notconnect 1         auto     auto    10/100BaseTX
Fa0/15    notconnect 1         auto     auto    10/100BaseTX
Fa0/16    notconnect 1         auto     auto    10/100BaseTX
Fa0/17    notconnect 1         auto     auto    10/100BaseTX
Fa0/18    notconnect 1         auto     auto    10/100BaseTX
Fa0/19    notconnect 1         auto     auto    10/100BaseTX
Fa0/20    notconnect 1         auto     auto    10/100BaseTX
Fa0/21    notconnect 1         auto     auto    10/100BaseTX
Fa0/22    notconnect 1         auto     auto    10/100BaseTX
Fa0/23    notconnect 1         auto     auto    10/100BaseTX
Fa0/24    notconnect 1         auto     auto    10/100BaseTX
Gig0/1    connected  trunk     a-full    a-100   10/100/1000BaseTX
Gig0/2    notconnect 1         auto     auto    10/100/1000BaseTX

SW-BR1#
```

SW-BR2

```
SW-BR2#sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     Name      notconnect  120       auto    auto   10/100BaseTX
Fa0/2     connected  120       a-full    a-100   10/100BaseTX
Fa0/3     notconnect 120       auto     auto    10/100BaseTX
Fa0/4     notconnect 120       auto     auto    10/100BaseTX
Fa0/5     notconnect 120       auto     auto    10/100BaseTX
Fa0/6     notconnect 120       auto     auto    10/100BaseTX
Fa0/7     notconnect 120       auto     auto    10/100BaseTX
Fa0/8     notconnect 120       auto     auto    10/100BaseTX
Fa0/9     notconnect 120       auto     auto    10/100BaseTX
Fa0/10    notconnect 120       auto     auto    10/100BaseTX
Fa0/11    notconnect 1         auto     auto    10/100BaseTX
Fa0/12    notconnect 1         auto     auto    10/100BaseTX
Fa0/13    notconnect 1         auto     auto    10/100BaseTX
Fa0/14    notconnect 1         auto     auto    10/100BaseTX
Fa0/15    notconnect 1         auto     auto    10/100BaseTX
Fa0/16    notconnect 1         auto     auto    10/100BaseTX
Fa0/17    notconnect 1         auto     auto    10/100BaseTX
Fa0/18    notconnect 1         auto     auto    10/100BaseTX
Fa0/19    notconnect 1         auto     auto    10/100BaseTX
Fa0/20    notconnect 1         auto     auto    10/100BaseTX
Fa0/21    notconnect 1         auto     auto    10/100BaseTX
Fa0/22    notconnect 1         auto     auto    10/100BaseTX
Fa0/23    notconnect 1         auto     auto    10/100BaseTX
Fa0/24    notconnect 1         auto     auto    10/100BaseTX
Gig0/1    connected  trunk     a-full    a-100   10/100/1000BaseTX
Gig0/2    notconnect 1         auto     auto    10/100/1000BaseTX

SW-BR2#
```

2. IP Addressing & Gateway Validation

Purpose: Confirm that the configured addresses match the project addressing plan.

Test / Evidence	Command or Action	Expected Result
Router IPs	show ip interface brief	Expected IP addresses are present
PC addressing	Desktop > IP Configuration	IP, mask and gateway are correct
Default gateway	ipconfig /all or PC IP Configuration	Gateway points to the intended router
Management IPs	show ip interface brief / VLAN interface	Management addressing is reachable

Evidence Screenshot

PC Addressing

1. Operation

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0001.63CC.7B86
    Link-local IPv6 Address.....: FE80::201:63FF:FECC:7B86
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.10.10
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.10.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-0A-DB-2B-A5-00-01-63-CC-7B-86
    DNS Servers.....: ::
                        192.168.50.10
```

2. Finance

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0060.2FB0.6C7E
    Link-local IPv6 Address.....: FE80::260:2FFF:FEB0:6C7E
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.20.10
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.20.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-CD-87-4E-31-00-60-2F-B0-6C-7E
    DNS Servers.....: ::
                        192.168.50.10
```

3. Customer Service

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 00E0.8FAE.643C
    Link-local IPv6 Address.....: FE80::2E0:8FFF:FEAE:643C
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.30.10
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.30.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-14-70-94-54-00-E0-8F-AE-64-3C
    DNS Servers.....: ::
                        192.168.50.10

```

4. IT Admin

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0060.7065.1078
    Link-local IPv6 Address.....: FE80::260:70FF:FE65:1078
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.40.10
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.40.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-C5-74-A3-DC-00-60-70-65-10-78
    DNS Servers.....: ::
                        192.168.50.10

```

5. Branch 1 PC

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0060.473C.9B5B
    Link-local IPv6 Address.....: FE80::260:47FF:FE3C:9B5B
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.110.10
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.110.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-E9-98-CB-E6-00-60-47-3C-9B-5B
    DNS Servers.....: ::
                        192.168.50.10

```

6. Branch 2 PC


```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0001.96C1.8680
    Link-local IPv6 Address.....: FE80::201:96FF:FEC1:8680
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.120.10
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        192.168.120.1
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-57-96-30-13-00-01-96-C1-86-80
    DNS Servers.....: ::
                        192.168.50.10

```

Switch

* Invalid input detected at ...

```
SW-H01#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Gig0/2
10	OPERATIONS	active	
20	FINANCE	active	
30	CUSTOMER_SERVICE	active	
40	IT_ADMIN	active	
50	SERVERS	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
99	MANAGEMENT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-H02#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/21, Fa0/22, Gig0/1
10	OPERATIONS	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
20	FINANCE	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10
30	CUSTOMER_SERVICE	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15
40	IT_ADMIN	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20
50	SERVERS	active	
99	MANAGEMENT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW-BR2# sh vlan

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
99 MANAGEMENT	active	
120 TELLER	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

VLAN Name: STP MTU: 1500 Name: Bridge-No: Bridge-No: Cap: Bridge-Md5: Trunk-Id: Trunk-Id:

SW-BR1# sh vlan

VLAN Name	Status	Ports
1 default	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
99 MANAGEMENT	active	
110 TELLER	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

3. Trunking & Inter-Switch Connectivity

Purpose: Verify trunk links between the routing/switching layers.

Test / Evidence	Command or Action	Expected Result
Trunk status	show interfaces trunk	Expected trunk ports are trunking
Allowed VLANs	show interfaces trunk	Required VLANs are allowed
Native VLAN	show interfaces trunk	Native VLAN matches the design

Evidence Screenshot

SW-HO1

```
-----
SW-HO1# sh int trunk
Port      Mode      Encapsulation  Status      Native vlan
Po1       on        802.1q         trunking    1
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Po1       10,20,30,40,50,99
Gig0/1    10,20,30,40,50,99

Port      Vlans allowed and active in management domain
Po1       10,20,30,40,50,99
Gig0/1    10,20,30,40,50,99

Port      Vlans in spanning tree forwarding state and not pruned
Po1       10,20,30,40,50,99
Gig0/1    10,20,30,40,50,99
SW-HO1#
```

SW-HO2

```
-----
SW-HO2# sh int trunk
Port      Mode      Encapsulation  Status      Native vlan
Po1       on        802.1q         trunking    1
Gig0/2    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Po1       10,20,30,40,50,99
Gig0/2    10,20,30,40,50,99

Port      Vlans allowed and active in management domain
Po1       10,20,30,40,50,99
Gig0/2    10,20,30,40,50,99

Port      Vlans in spanning tree forwarding state and not pruned
Po1       10,20,30,40,50,99
Gig0/2    10,20,30,40,50,99
SW-HO2#
```

4. Inter-VLAN Routing & Ping Matrix

Purpose: Prove that routing between permitted internal VLANs works.

Test / Evidence	Command or Action	Expected Result
Operations → Finance	ping Finance host from Operations host	Reply received if policy permits
Operations → Customer Service	ping Customer Service host	Reply received if policy permits
IT Admin → Server	ping banking server	Reply received

Evidence Screenshot

Operations to Finance

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Operations to Customer Service

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

IT Admin to Server

```
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\> ping 192.168.50.10

Pinging 192.168.50.10 with 32 bytes of data:

Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time=1ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

IT Admin to Branch 1 and Branch 2

```
C:\>ping 192.168.120.10

Pinging 192.168.120.10 with 32 bytes of data:

Reply from 192.168.120.10: bytes=32 time=1ms TTL=126
Reply from 192.168.120.10: bytes=32 time=2ms TTL=126
Reply from 192.168.120.10: bytes=32 time=1ms TTL=126
Reply from 192.168.120.10: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.120.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 192.168.110.10

Pinging 192.168.110.10 with 32 bytes of data:

Reply from 192.168.110.10: bytes=32 time=2ms TTL=126
Reply from 192.168.110.10: bytes=32 time=1ms TTL=126
Reply from 192.168.110.10: bytes=32 time=23ms TTL=126
Reply from 192.168.110.10: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.110.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 23ms, Average = 7ms

C:\>|
```

Finance to IT Admin

```
C:\> ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127
Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

5. WAN Connectivity & Branch Reachability

Purpose: Verify Head Office to Branch 1 and Branch 2 connectivity.

Test / Evidence	Command or Action	Expected Result
HO → Branch 1	ping Branch 1 router/teller	Successful replies
HO → Branch 2	ping Branch 2 router/teller	Successful replies
Branch 1 → HO	ping an HO server or management address	Successful replies
Branch 2 → HO	ping an HO server or management address	Successful replies

Evidence Screenshot

HO1 -Branch1

```
Passw0rd:
R-HO1#ping 192.168.110.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.110.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/15/34 ms

R-HO1#
```

HO2 -Branch2

```
R-HO1#ping 192.168.120.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.120.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/11/20 ms

R-HO1#
```

Branch1 - HO

```
C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time=20ms TTL=254
Reply from 10.0.0.1: bytes=32 time=1ms TTL=254
Reply from 10.0.0.1: bytes=32 time=1ms TTL=254
Reply from 10.0.0.1: bytes=32 time=38ms TTL=254

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 38ms, Average = 15ms
```

Branch2 -H0

```
C:\> ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time=10ms TTL=254
Reply from 10.0.0.1: bytes=32 time=1ms TTL=254
Reply from 10.0.0.1: bytes=32 time=1ms TTL=254
Reply from 10.0.0.1: bytes=32 time=24ms TTL=254

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 24ms, Average = 9ms

C:\>
```

6. OSPF Neighbor & Routing Validation

Purpose: Verify dynamic WAN routing and route learning.

Test / Evidence	Command or Action	Expected Result
OSPF neighbors	show ip ospf neighbor	Expected neighbors reach Full state
OSPF routes	show ip route ospf	Remote networks appear as O routes
OSPF process	show ip protocols	Correct OSPF process/networks are shown

Evidence Screenshot

R-HO1

```
R-HO1#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	1	FULL/DR	00:00:36	192.168.30.3	GigabitEthernet0/0.30
2.2.2.2	1	FULL/DR	00:00:37	192.168.40.3	GigabitEthernet0/0.40
2.2.2.2	1	FULL/DR	00:00:37	192.168.50.3	GigabitEthernet0/0.50
2.2.2.2	1	FULL/DR	00:00:36	192.168.99.3	GigabitEthernet0/0.99
4.4.4.4	0	FULL/ -	00:00:35	10.0.0.2	Serial0/0/1
2.2.2.2	1	FULL/DR	00:00:37	192.168.10.3	GigabitEthernet0/0.10
2.2.2.2	1	FULL/DR	00:00:37	192.168.20.3	GigabitEthernet0/0.20
3.3.3.3	0	FULL/ -	00:00:36	10.0.1.2	Serial0/0/0

```
R-HO1#
```

```
R-HO1#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 20.0.0.2 to network 0.0.0.0

```
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial0/0/1
L    10.0.0.1/32 is directly connected, Serial0/0/1
C    10.0.1.0/30 is directly connected, Serial0/0/0
L    10.0.1.1/32 is directly connected, Serial0/0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/30 is directly connected, GigabitEthernet0/1
L    20.0.0.1/32 is directly connected, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.10.2/32 is directly connected, GigabitEthernet0/0.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.20.2/32 is directly connected, GigabitEthernet0/0.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.30.2/32 is directly connected, GigabitEthernet0/0.30
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L    192.168.40.2/32 is directly connected, GigabitEthernet0/0.40
192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L    192.168.50.2/32 is directly connected, GigabitEthernet0/0.50
192.168.99.0/24 is variably subnetted, 4 subnets, 2 masks
C    192.168.99.0/28 is directly connected, GigabitEthernet0/0.99
L    192.168.99.2/32 is directly connected, GigabitEthernet0/0.99
O    192.168.99.16/28 [110/65] via 10.0.1.2, 4294967274:4294967255:4294967250, Serial0/0/0
O    192.168.99.32/28 [110/65] via 10.0.0.2, 4294967274:4294967255:4294967250, Serial0/0/1
O    192.168.110.0/24 [110/65] via 10.0.1.2, 4294967274:4294967255:4294967250, Serial0/0/0
O    192.168.120.0/24 [110/65] via 10.0.0.2, 4294967274:4294967255:4294967250, Serial0/0/1
S*   0.0.0.0/0 [1/0] via 20.0.0.2
```



```

R-H01#sh ip protocols

Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    192.168.50.0 0.0.0.255 area 0
    10.0.1.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
    192.168.99.0 0.0.0.15 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:17:04
    2.2.2.2          110          00:17:02
    3.3.3.3          110          00:17:49
    4.4.4.4          110          00:17:47
  Distance: (default is 110)

R-H01#

```

R-H02

```
R-H02# sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
1.1.1.1	1	FULL/BDR	00:00:31	192.168.10.2	GigabitEthernet0/0.10
1.1.1.1	1	FULL/BDR	00:00:31	192.168.20.2	GigabitEthernet0/0.20
1.1.1.1	1	FULL/BDR	00:00:31	192.168.30.2	GigabitEthernet0/0.30
1.1.1.1	1	FULL/BDR	00:00:31	192.168.40.2	GigabitEthernet0/0.40
1.1.1.1	1	FULL/BDR	00:00:31	192.168.50.2	GigabitEthernet0/0.50
1.1.1.1	1	FULL/BDR	00:00:30	192.168.99.2	GigabitEthernet0/0.99

```

R-H02#

```

```

R-H02# sh ip protocols

Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.10.0 0.0.0.255 area 0
    192.168.20.0 0.0.0.255 area 0
    192.168.30.0 0.0.0.255 area 0
    192.168.40.0 0.0.0.255 area 0
    192.168.50.0 0.0.0.255 area 0
    192.168.99.0 0.0.0.15 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:17:24
    2.2.2.2          110          00:17:21
    3.3.3.3          110          00:18:09
    4.4.4.4          110          00:18:07
  Distance: (default is 110)

```

R-H02# sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 20.0.0.6 to network 0.0.0.0

```
10.0.0.0/30 is subnetted, 2 subnets
O    10.0.0.0/30 [110/65] via 192.168.10.2, 01:18:24, GigabitEthernet0/0.10
      [110/65] via 192.168.20.2, 01:18:24, GigabitEthernet0/0.20
      [110/65] via 192.168.30.2, 01:18:24, GigabitEthernet0/0.30
      [110/65] via 192.168.40.2, 01:18:24, GigabitEthernet0/0.40
      [110/65] via 192.168.50.2, 01:18:24, GigabitEthernet0/0.50
      [110/65] via 192.168.99.2, 01:18:24, GigabitEthernet0/0.99
O    10.0.1.0/30 [110/65] via 192.168.10.2, 01:18:24, GigabitEthernet0/0.10
      [110/65] via 192.168.20.2, 01:18:24, GigabitEthernet0/0.20
      [110/65] via 192.168.30.2, 01:18:24, GigabitEthernet0/0.30
      [110/65] via 192.168.40.2, 01:18:24, GigabitEthernet0/0.40
      [110/65] via 192.168.50.2, 01:18:24, GigabitEthernet0/0.50
      [110/65] via 192.168.99.2, 01:18:24, GigabitEthernet0/0.99
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.4/30 is directly connected, GigabitEthernet0/1
L    20.0.0.5/32 is directly connected, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.10.3/32 is directly connected, GigabitEthernet0/0.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.20.3/32 is directly connected, GigabitEthernet0/0.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.30.3/32 is directly connected, GigabitEthernet0/0.30
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L    192.168.40.3/32 is directly connected, GigabitEthernet0/0.40
192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L    192.168.50.3/32 is directly connected, GigabitEthernet0/0.50
192.168.99.0/24 is variably subnetted, 4 subnets, 2 masks
C    192.168.99.0/28 is directly connected, GigabitEthernet0/0.99
L    192.168.99.3/32 is directly connected, GigabitEthernet0/0.99
O    192.168.99.16/28 [110/66] via 192.168.10.2, 01:18:24, GigabitEthernet0/0.10
      [110/66] via 192.168.20.2, 01:18:24, GigabitEthernet0/0.20
      [110/66] via 192.168.30.2, 01:18:24, GigabitEthernet0/0.30
      [110/66] via 192.168.40.2, 01:18:24, GigabitEthernet0/0.40
      [110/66] via 192.168.50.2, 01:18:24, GigabitEthernet0/0.50
      [110/66] via 192.168.99.2, 01:18:24, GigabitEthernet0/0.99
O    192.168.99.32/28 [110/66] via 192.168.10.2, 01:18:24, GigabitEthernet0/0.10
      [110/66] via 192.168.20.2, 01:18:24, GigabitEthernet0/0.20
```

R-BR2

```
R-BR2# sh ip protocols

Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.4.4.4
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    192.168.120.0 0.0.0.255 area 0
    192.168.99.32 0.0.0.15 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:18:12
    2.2.2.2          110          00:18:09
    3.3.3.3          110          00:18:57
    4.4.4.4          110          00:18:55
  Distance: (default is 110)
```

```
R-BR2#
```

R-BR1

```
R-BR1# sh ip protocols

Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.110.0 0.0.0.255 area 0
    10.0.1.0 0.0.0.3 area 0
    192.168.99.16 0.0.0.15 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:18:56
    2.2.2.2          110          00:18:54
    3.3.3.3          110          00:19:41
    4.4.4.4          110          00:19:40
  Distance: (default is 110)
```

```
R-BR1#
```

```
R-BR1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O       10.0.0.0/30 [110/128] via 10.0.1.1, 01:20:40, Serial0/0/0
C       10.0.1.0/30 is directly connected, Serial0/0/0
L       10.0.1.2/32 is directly connected, Serial0/0/0
O       192.168.10.0/24 [110/65] via 10.0.1.1, 01:19:55, Serial0/0/0
O       192.168.20.0/24 [110/65] via 10.0.1.1, 01:19:55, Serial0/0/0
O       192.168.30.0/24 [110/65] via 10.0.1.1, 01:20:05, Serial0/0/0
O       192.168.40.0/24 [110/65] via 10.0.1.1, 01:19:55, Serial0/0/0
O       192.168.50.0/24 [110/65] via 10.0.1.1, 01:19:55, Serial0/0/0
    192.168.99.0/24 is variably subnetted, 4 subnets, 2 masks
O       192.168.99.0/28 [110/65] via 10.0.1.1, 01:19:55, Serial0/0/0
C       192.168.99.16/28 is directly connected, GigabitEthernet0/0.99
L       192.168.99.17/32 is directly connected, GigabitEthernet0/0.99
O       192.168.99.32/28 [110/129] via 10.0.1.1, 01:20:30, Serial0/0/0
    192.168.110.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.110.0/24 is directly connected, GigabitEthernet0/0.110
L       192.168.110.1/32 is directly connected, GigabitEthernet0/0.110
O       192.168.120.0/24 [110/129] via 10.0.1.1, 01:20:30, Serial0/0/0
```

```
R-BR1#
```

7. HSRP Gateway Redundancy & Failover

Purpose: Prove that the primary/standby gateway design survives router failure.

Test / Evidence	Command or Action	Expected Result
HSRP state	show standby brief	R-HO1 Active and R-HO2 Standby as designed
Virtual gateway	Check PC default gateway	Users use the virtual gateway
Failure test	Shut down the active HSRP interface/router	R-HO2 becomes Active
Recovery	Restore the failed router/interface	Expected HSRP roles return according to priority/preempt design

Evidence Screenshot

HSRP before Failure

```
R-HO1#sh standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State   Active        Standby        Virtual IP
Gig        10   120 P Active   local         192.168.10.3   192.168.10.1
Gig        20   120 P Active   local         192.168.20.3   192.168.20.1
Gig        30   120 P Active   local         192.168.30.3   192.168.30.1
Gig        40   120 P Active   local         192.168.40.3   192.168.40.1
Gig        50   120 P Active   local         192.168.50.3   192.168.50.1
Gig        99   120 P Active   local         192.168.99.3   192.168.99.1
R-HO1#
```

```
R-HO2#sh standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri P State   Active        Standby        Virtual IP
Gig        10   100 P Standby  192.168.10.2   local         192.168.10.1
Gig        20   100 P Standby  192.168.20.2   local         192.168.20.1
Gig        30   100 P Standby  192.168.30.2   local         192.168.30.1
Gig        40   100 P Standby  192.168.40.2   local         192.168.40.1
Gig        50   100 P Standby  192.168.50.2   local         192.168.50.1
Gig        99   100 P Standby  192.168.99.2   local         192.168.99.1
R-HO2#
```

Operations PC Connectivity to HSRP Virtual Gateway

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

HSRP After Failure

```
administratively down down
R-HO1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R-HO1(config)#int g0/0.10
R-HO1(config-subif)#shut

R-HO1(config-subif)#
%HSRP-6-STATECHANGE: GigabitEthernet0/0.10 Grp 10 state
Active -> Init

%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed
state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0.10, changed state to down

00:55:20: %OSPF-5-ADJCHG: Process 100, Nbr 2.2.2.2 on
GigabitEthernet0/0.10 from FULL to DOWN, Neighbor Down:
Interface down or detached
```

HSRP Failover: R-HO2 Becomes Active

```
R-HO2#sh standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State    Active        Standby        Virtual IP
Gig        10   100  P Active   local         192.168.10.3   192.168.10.1
Gig        20   100  P Standby  192.168.20.2  local          192.168.20.1
Gig        30   100  P Standby  192.168.30.2  local          192.168.30.1
Gig        40   100  P Standby  192.168.40.2  local          192.168.40.1
Gig        50   100  P Standby  192.168.50.2  local          192.168.50.1
Gig        99   100  P Standby  192.168.99.2  local          192.168.99.1
R-HO2#
```

Successful Connectivity After HSRP Failover

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

HSRP Recovery

```
R-HO1(config-subif)#int g0/0.10
R-HO1(config-subif)#no shut

R-HO1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%HSRP-6-STATECHANGE: GigabitEthernet0/0.10 Grp 10 state Standby -> Active
```

```
R-HO1#sh standby brief
          P indicates configured to preempt.
          |
Interface    Grp  Pri P State    Active            Standby            Virtual IP
Gig          10   120 P Active   local             192.168.10.3       192.168.10.1
Gig          20   120 P Active   local             192.168.20.3       192.168.20.1
Gig          30   120 P Active   local             192.168.30.3       192.168.30.1
Gig          40   120 P Active   local             192.168.40.3       192.168.40.1
Gig          50   120 P Active   local             192.168.50.3       192.168.50.1
Gig          99   120 P Active   local             192.168.99.3       192.168.99.1
R-HO1#
```

8. Default Route & PAT / Internet Access

Purpose: Validate external connectivity through the simulated ISP.

Test / Evidence	Command or Action	Expected Result
Default route	show ip route	Default route points toward the simulated ISP
PAT translations	show ip nat translations	Active translations appear after traffic
PAT statistics	show ip nat statistics	Inside/outside interfaces and translation statistics are shown
External ping	Ping the simulated external destination	Successful reply if the external test target is configured

Evidence Screenshot

```
Gig          99    120 P Active   local          192.168.99.3    192.168.99.1
R-H01#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 20.0.0.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/1
L       10.0.0.1/32 is directly connected, Serial0/0/1
C       10.0.1.0/30 is directly connected, Serial0/0/0
L       10.0.1.1/32 is directly connected, Serial0/0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/30 is directly connected, GigabitEthernet0/1
L       20.0.0.1/32 is directly connected, GigabitEthernet0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.2/32 is directly connected, GigabitEthernet0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.2/32 is directly connected, GigabitEthernet0/0.20
    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.2/32 is directly connected, GigabitEthernet0/0.30
    192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.40.2/32 is directly connected, GigabitEthernet0/0.40
    192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L       192.168.50.2/32 is directly connected, GigabitEthernet0/0.50
    192.168.99.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.99.0/28 is directly connected, GigabitEthernet0/0.99
L       192.168.99.2/32 is directly connected, GigabitEthernet0/0.99
O       192.168.99.16/28 [110/65] via 10.0.1.2, 4294967274:4294967271:4294967271, Serial0/0/0
O       192.168.99.32/28 [110/65] via 10.0.0.2, 4294967274:4294967271:4294967271, Serial0/0/1
O       192.168.110.0/24 [110/65] via 10.0.1.2, 4294967274:4294967271:4294967271, Serial0/0/0
O       192.168.120.0/24 [110/65] via 10.0.0.2, 4294967274:4294967271:4294967271, Serial0/0/1
S*     0.0.0.0/0 [1/0] via 20.0.0.2
```



```

C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=24ms TTL=254
Reply from 20.0.0.2: bytes=32 time<1ms TTL=254
Reply from 20.0.0.2: bytes=32 time<1ms TTL=254
Reply from 20.0.0.2: bytes=32 time<1ms TTL=254

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 24ms, Average = 6ms

C:\>

```

```

R-H01# sh ip nat statistics
Total translations: 16 (0 static, 16 dynamic, 16 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/0.10 , GigabitEthernet0/0.20 , GigabitEthernet0/0.30 ,
Hits: 27 Misses: 28
Expired translations: 12
Dynamic mappings:
R-H01#

```

```

R-H01#sh ip nat translations
R-H01#sh ip nat translations

```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	20.0.0.1:13	192.168.20.10:13	20.0.0.2:13	20.0.0.2:13
icmp	20.0.0.1:14	192.168.20.10:14	20.0.0.2:14	20.0.0.2:14
icmp	20.0.0.1:15	192.168.20.10:15	20.0.0.2:15	20.0.0.2:15
icmp	20.0.0.1:16	192.168.20.10:16	20.0.0.2:16	20.0.0.2:16
icmp	20.0.0.1:17	192.168.20.10:17	20.0.0.6:17	20.0.0.6:17
icmp	20.0.0.1:18	192.168.20.10:18	20.0.0.6:18	20.0.0.6:18
icmp	20.0.0.1:19	192.168.20.10:19	20.0.0.6:19	20.0.0.6:19
icmp	20.0.0.1:20	192.168.20.10:20	20.0.0.6:20	20.0.0.6:20
icmp	20.0.0.1:53	192.168.10.10:53	20.0.0.6:53	20.0.0.6:53
icmp	20.0.0.1:54	192.168.10.10:54	20.0.0.6:54	20.0.0.6:54
icmp	20.0.0.1:55	192.168.10.10:55	20.0.0.6:55	20.0.0.6:55
icmp	20.0.0.1:56	192.168.10.10:56	20.0.0.6:56	20.0.0.6:56
icmp	20.0.0.1:57	192.168.10.10:57	20.0.0.2:57	20.0.0.2:57
icmp	20.0.0.1:58	192.168.10.10:58	20.0.0.2:58	20.0.0.2:58
icmp	20.0.0.1:59	192.168.10.10:59	20.0.0.2:59	20.0.0.2:59
icmp	20.0.0.1:60	192.168.10.10:60	20.0.0.2:60	20.0.0.2:60
icmp	20.0.0.1:61	192.168.10.10:61	20.0.0.6:61	20.0.0.6:61
icmp	20.0.0.1:62	192.168.10.10:62	20.0.0.6:62	20.0.0.6:62
icmp	20.0.0.1:63	192.168.10.10:63	20.0.0.6:63	20.0.0.6:63
icmp	20.0.0.1:64	192.168.10.10:64	20.0.0.6:64	20.0.0.6:64

9. ACL Security Validation

Purpose: Test both permitted and denied traffic and prove that the ACL is enforcing the intended policy.

Test / Evidence	Command or Action	Expected Result
ACL configuration	show access-lists	Correct entries and sequence are present
Allowed traffic	Ping/access a permitted destination	Traffic succeeds
Denied traffic	Ping/access a restricted destination	Traffic fails as intended

Evidence Screenshot

R-HO1

```
R-HO1#sh access-lists
Standard IP access list 1
 10 permit 192.168.0.0 0.0.255.255 (56 match(es))
Standard IP access list 10
 10 permit 192.168.40.0 0.0.0.255
 20 deny any
Extended IP access list operations
 10 permit tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq www
 20 permit udp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq domain
 30 permit udp 192.168.10.0 0.0.0.255 host 192.168.50.20 eq 123
 40 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 (8 match(es))
 50 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255 (4 match(es))
 60 deny ip 192.168.10.0 0.0.0.255 192.168.99.0 0.0.0.255
 70 permit ip any any (6031 match(es))
Extended IP access list isp
 10 permit tcp any any
 20 permit icmp any any echo-reply (27 match(es))
 30 deny ip 127.0.0.0 0.255.255.255 any
 40 deny ip any any
Extended IP access list finance
 10 permit tcp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq www
 20 permit udp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq domain
 30 permit udp 192.168.20.0 0.0.0.255 host 192.168.50.20 eq 123
 40 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 50 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 60 deny ip 192.168.20.0 0.0.0.255 192.168.99.0 0.0.0.255
 70 permit ip any any (2934 match(es))
Extended IP access list customer
 10 permit tcp 192.168.30.0 0.0.0.255 host 192.168.50.10 eq www
 20 permit udp 192.168.30.0 0.0.0.255 host 192.168.50.10 eq domain
 30 permit udp 192.168.30.0 0.0.0.255 host 192.168.50.20 eq 123
 40 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 50 deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255 (4 match(es))
 60 deny ip 192.168.30.0 0.0.0.255 192.168.99.0 0.0.0.255
 70 permit ip any any (6086 match(es))
Extended IP access list itadmin
 10 permit ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255 (7 match(es))
 20 permit ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255 (4 match(es))
 30 permit ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255 (4 match(es))
 40 permit ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255 (20 match(es))
 50 permit ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
 60 permit ip 192.168.40.0 0.0.0.255 192.168.110.0 0.0.0.255 (4 match(es))
 70 permit ip 192.168.40.0 0.0.0.255 192.168.120.0 0.0.0.255 (4 match(es))
 80 permit ip any any (6084 match(es))
Extended IP access list management
 10 permit ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
 20 permit ip 192.168.99.0 0.0.0.255 any (6087 match(es))
 30 deny ip any 192.168.99.0 0.0.0.255
 40 permit ip any any
Extended IP access list bankserver
 10 permit udp 192.168.110.0 0.0.0.255 host 192.168.50.10 eq bootps
 20 permit udp 192.168.120.0 0.0.0.255 host 192.168.50.10 eq bootps
 30 permit tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq www
 40 permit tcp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq www
```

```

70 permit ip any any (6086 match(es))
Extended IP access list itadmin
10 permit ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255 (7 match(es))
20 permit ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255 (4 match(es))
30 permit ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255 (4 match(es))
40 permit ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255 (20 match(es))
50 permit ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
60 permit ip 192.168.40.0 0.0.0.255 192.168.110.0 0.0.0.255 (4 match(es))
70 permit ip 192.168.40.0 0.0.0.255 192.168.120.0 0.0.0.255 (4 match(es))
80 permit ip any any (6084 match(es))
Extended IP access list management
10 permit ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
20 permit ip 192.168.99.0 0.0.0.255 any (6087 match(es))
30 deny ip any 192.168.99.0 0.0.0.255
40 permit ip any any
Extended IP access list bankserver
10 permit udp 192.168.110.0 0.0.0.255 host 192.168.50.10 eq bootps
20 permit udp 192.168.120.0 0.0.0.255 host 192.168.50.10 eq bootps
30 permit tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq www
40 permit tcp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq www
50 permit tcp 192.168.30.0 0.0.0.255 host 192.168.50.10 eq www
60 permit tcp 192.168.110.0 0.0.0.255 host 192.168.50.10 eq www
70 permit tcp 192.168.120.0 0.0.0.255 host 192.168.50.10 eq www
80 permit ip 192.168.40.0 0.0.0.255 host 192.168.50.10 (8 match(es))
90 permit tcp host 192.168.50.10 192.168.10.0 0.0.0.255 established
100 permit tcp host 192.168.50.10 192.168.20.0 0.0.0.255 established
110 permit tcp host 192.168.50.10 192.168.30.0 0.0.0.255 established
120 permit tcp host 192.168.50.10 192.168.110.0 0.0.0.255 established
130 permit tcp host 192.168.50.10 192.168.120.0 0.0.0.255 established
140 permit udp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq domain
150 permit udp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq domain
160 permit udp 192.168.30.0 0.0.0.255 host 192.168.50.10 eq domain
170 permit udp 192.168.110.0 0.0.0.255 host 192.168.50.10 eq domain
180 permit udp 192.168.120.0 0.0.0.255 host 192.168.50.10 eq domain
190 deny ip any any (12 match(es))

```

R-H01#

R - HO2

```

R-HO2#sh access-lists
Standard IP access list 1
 10 permit 192.168.0.0 0.0.255.255
Standard IP access list 10
 10 permit 192.168.40.0 0.0.0.255
 20 deny any
Extended IP access list operations
 10 permit tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq www
 20 permit udp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq domain
 30 permit udp 192.168.10.0 0.0.0.255 host 192.168.50.20 eq 123
 40 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 50 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
 60 deny ip 192.168.10.0 0.0.0.255 192.168.99.0 0.0.0.255
 70 permit ip any any (6017 match(es))
Extended IP access list isp
 10 permit tcp any any
 20 permit icmp any any echo-reply
 30 deny ip 127.0.0.0 0.255.255.255 any
 40 deny ip any any
Extended IP access list finance
 10 permit tcp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq www
 20 permit udp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq domain
 30 permit udp 192.168.20.0 0.0.0.255 host 192.168.50.20 eq 123
 40 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 50 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 60 deny ip 192.168.20.0 0.0.0.255 192.168.99.0 0.0.0.255
 70 permit ip any any (2940 match(es))
Extended IP access list customer
 10 permit tcp 192.168.30.0 0.0.0.255 host 192.168.50.10 eq www
 20 permit udp 192.168.30.0 0.0.0.255 host 192.168.50.10 eq domain
 30 permit udp 192.168.30.0 0.0.0.255 host 192.168.50.20 eq 123
 40 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 50 deny ip 192.168.30.0 0.0.0.255 192.168.40.0 0.0.0.255
 60 deny ip 192.168.30.0 0.0.0.255 192.168.99.0 0.0.0.255
 70 permit ip any any (6109 match(es))
Extended IP access list itadmin
 10 permit ip 192.168.40.0 0.0.0.255 192.168.10.0 0.0.0.255
 20 permit ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
 30 permit ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
 40 permit ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
 50 permit ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
 60 permit ip 192.168.40.0 0.0.0.255 192.168.110.0 0.0.0.255
 70 permit ip 192.168.40.0 0.0.0.255 192.168.120.0 0.0.0.255
 80 permit ip any any (6111 match(es))
Extended IP access list management
 10 permit ip 192.168.40.0 0.0.0.255 192.168.99.0 0.0.0.255
 20 permit ip 192.168.99.0 0.0.0.255 any (6106 match(es))
 30 deny ip any 192.168.99.0 0.0.0.255
 40 permit ip any any
Extended IP access list bankserver
 10 permit udp 192.168.110.0 0.0.0.255 host 192.168.50.10 eq bootps
 20 permit udp 192.168.120.0 0.0.0.255 host 192.168.50.10 eq bootps
 30 permit tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq www
 40 permit tcp 192.168.20.0 0.0.0.255 host 192.168.50.10 eq www

```

R-BR1

```

R-BR1# sh access-lists
Standard IP access list 10
 10 permit 192.168.40.0 0.0.0.255
 20 deny any
Extended IP access list branch1
 10 permit tcp 192.168.110.0 0.0.0.255 host 192.168.50.10
 20 permit tcp 192.168.110.0 0.0.0.255 host 192.168.50.10 eq www
 30 permit udp 192.168.110.0 0.0.0.255 host 192.168.50.10 eq domain
 40 permit udp 192.168.110.0 0.0.0.255 host 192.168.50.20 eq 123
 50 deny ip 192.168.110.0 0.0.0.255 192.168.10.0 0.0.0.255 (12 match(es))
 60 deny ip 192.168.110.0 0.0.0.255 192.168.20.0 0.0.0.255 (4 match(es))
 70 deny ip 192.168.110.0 0.0.0.255 192.168.30.0 0.0.0.255
 80 deny ip 192.168.110.0 0.0.0.255 192.168.99.0 0.0.0.255
 90 deny ip 192.168.110.0 0.0.0.255 192.168.120.0 0.0.0.255
 100 permit ip any any (17 match(es))

R-BR1#

```

R-BR2

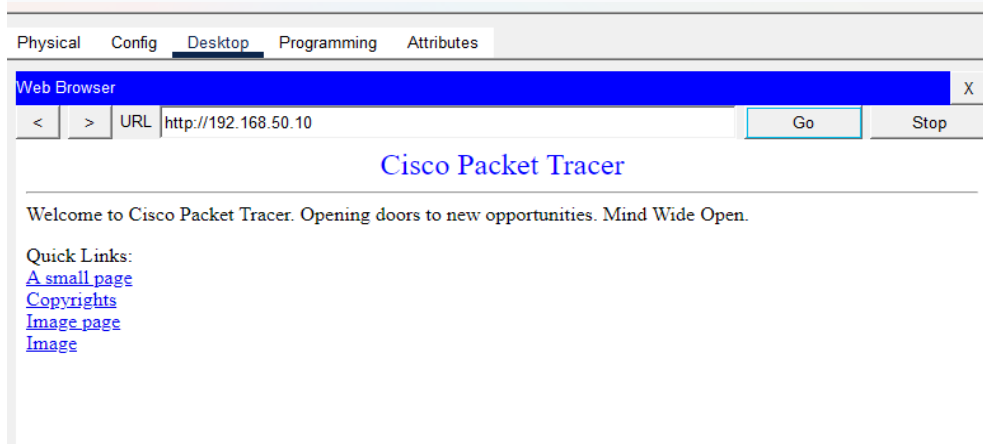
```
R-BR2#sh access-lists
Standard IP access list 10
 10 permit 192.168.40.0 0.0.0.255
 20 deny any
Extended IP access list branch2
 10 permit tcp 192.168.120.0 0.0.0.255 host 192.168.50.10 eq www
 20 permit udp 192.168.120.0 0.0.0.255 host 192.168.50.10 eq domain
 30 permit udp 192.168.120.0 0.0.0.255 host 192.168.50.20 eq 123
 40 deny ip 192.168.120.0 0.0.0.255 192.168.10.0 0.0.0.255 (12 match(es))
 50 deny ip 192.168.120.0 0.0.0.255 192.168.20.0 0.0.0.255 (4 match(es))
 60 deny ip 192.168.120.0 0.0.0.255 192.168.30.0 0.0.0.255
 70 deny ip 192.168.120.0 0.0.0.255 192.168.99.0 0.0.0.255
 80 deny ip 192.168.120.0 0.0.0.255 192.168.110.0 0.0.0.255 (4 match(es))
 90 permit ip any any (17 match(es))

R-BR2#
```

Purpose: Test both permitted and denied traffic and prove that the ACL is enforcing the intended policy.

```
ip access-list extended operations
permit tcp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq www
permit udp 192.168.10.0 0.0.0.255 host 192.168.50.10 eq domain
permit udp 192.168.10.0 0.0.0.255 host 192.168.50.20 eq 123
deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
deny ip 192.168.10.0 0.0.0.255 192.168.99.0 0.0.0.255
permit ip any any
```

Evidence Screenshot



Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping bank.com

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time=1ms TTL=255
Reply from 20.0.0.1: bytes=32 time=1ms TTL=255
Reply from 20.0.0.1: bytes=32 time=9ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping bank.com

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time=1ms TTL=255
Reply from 20.0.0.1: bytes=32 time=1ms TTL=255
Reply from 20.0.0.1: bytes=32 time=9ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>nslookup bank.com

Server: [192.168.50.10]
Address: 192.168.50.10

Non-authoritative answer:
Name: bank.com
Address: 20.0.0.1

C:\>
```

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Reply from 192.168.10.2: Destination host unreachable.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.99.10

Pinging 192.168.99.10 with 32 bytes of data:

Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.
Reply from 192.168.10.2: Destination host unreachable.

Ping statistics for 192.168.99.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

10. SSH Management Security

Purpose: Verify that network-device management is available through SSH and not insecure remote access.

Test / Evidence	Command or Action	Expected Result
SSH status	show ip ssh	SSH is enabled
VTY configuration	show running-config section line vty	VTY uses SSH/login controls
SSH login	PC > Command Prompt > ssh ...	Authorized user can log in
Unauthorized test	Attempt invalid credentials	Access is rejected

Evidence Screenshot

```
R-H01#sh ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R-H01#
```

```
R-H01#sh run | section line vty
line vty 0 4
  access-class 10 in
  login local
  transport input ssh
line vty 5
  access-class 10 in
  login local
  transport input ssh
R-H01#
```


Successful SSH Login

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.99.2

Pinging 192.168.99.2 with 32 bytes of data:

Reply from 192.168.99.2: bytes=32 time<1ms TTL=255
Reply from 192.168.99.2: bytes=32 time<1ms TTL=255
Reply from 192.168.99.2: bytes=32 time=1ms TTL=255
Reply from 192.168.99.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.99.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ssh -l admin 192.168.99.2

Password:
% Login invalid

Password:

*****
*   Secure Banking Network - Head Office   *
*   Unauthorized Access is Prohibited      *
*****

R-H01>en
Password:
Password:
R-H01#sh ip int brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0        unassigned      YES manual  up          up
GigabitEthernet0/0.10     192.168.10.2    YES manual  up          up
GigabitEthernet0/0.20     192.168.20.2    YES manual  up          up
GigabitEthernet0/0.30     192.168.30.2    YES manual  up          up
GigabitEthernet0/0.40     192.168.40.2    YES manual  up          up
GigabitEthernet0/0.50     192.168.50.2    YES manual  up          up
GigabitEthernet0/0.99     192.168.99.2    YES manual  up          up
GigabitEthernet0/1        20.0.0.1        YES manual  up          up
GigabitEthernet0/2        unassigned      YES unset   administratively down down
Serial0/0/0               10.0.1.1        YES manual  up          up
Serial0/0/1               10.0.0.1        YES manual  up          up
Vlan1                     unassigned      YES unset   administratively down down
R-H01#
```

11. TFTP Backup & NTP Validation

Purpose: Validate management services used for configuration backup and time synchronization.

Test / Evidence	Command or Action	Expected Result
TFTP backup	copy running-config tftp:	Configuration backup completes
Backup file	Check TFTP server file list	Backup file exists
NTP status	show clock / show ntp status	Device time is synchronized as designed

Evidence Screenshot

TFTP Backup Files

The screenshot shows a network device configuration interface. The 'Services' tab is active. On the left, a list of services includes HTTP, DHCP, DHCPv6, TFTP (highlighted), DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP, and PRP. On the right, the TFTP service is configured. The 'Service' status is 'On' (indicated by a blue dot). Below this, a list of files is shown, including R-HO1-config, R-HO2-config, asa842-k8.bin, asa923-k8.bin, c1841-advipservicesk9-mz.124-15.T1.bin, c1841-ipbase-mz.123-14.T7.bin, c1841-ipbasek9-mz.124-12.bin, c1900-universalk9-mz.SPA.155-3.M4a.bin, c2600-advipservicesk9-mz.124-15.T1.bin, and c2600-i-mz.122-28.bin.

NTP Status

```
R-HO1#sh clock
0:28:52.379 UTC Sun Aug 9 2026
R-HO1#sh ntp status
Clock is synchronized, stratum 2, reference is 192.168.50.20
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is EDF6C6C5.00000082 (0:28:53.130 UTC Sun Aug 9 2026)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 65.70 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll interval is 4,
last update was 3 sec ago.
R-HO1#
```

Troubleshooting Command Cheat Sheet

Area	Command	What to Check
Interface / link	show ip interface brief	Check status and protocol
Neighbor discovery	show cdp neighbors	Confirm physical/logical neighbor and port
VLAN	show vlan brief	Confirm VLAN exists and access-port membership
Trunk	show interfaces trunk	Confirm trunk state and allowed VLANs
Routing	show ip route	Check whether destination network is known
OSPF	show ip ospf neighbor	Check OSPF adjacency
OSPF routes	show ip route ospf	Check learned OSPF routes
HSRP	show standby brief	Check Active/Standby state
NAT	show ip nat translations	Check PAT translations
ACL	show access-lists	Check rules and match counters
SSH	show ip ssh	Check SSH capability
Config	show running-config	Find incorrect configuration